CYBERSECURITY FOUNDATIONS · INSTRUCTOR COPY

Speaker Notes for All 23 Slides

These notes walk you through every slide of the deck: a suggested script, the one point that must land, a question to pull the class in, live demos where they pay off, and the misconceptions to watch for. The script is a starting point, not a cage — once the room is with you, improvise. The same notes are also embedded in the presenter view of the file “DS.01.01 - I.T. Fundamentals (with speaker notes).pptx”.

How each slide block works

Field	What it gives you
SAY	A suggested script in plain classroom language — say it your way.
LAND THIS	The one thing students must still remember tomorrow.
ASK THE CLASS	A question or poll to keep the room awake, with expected answers.
DO LIVE	A demo or board moment worth the setup time.
WATCH FOR	The misconception or confusion that reliably shows up.
TRANSITION	A one-line bridge into the next slide (unit dividers).

Suggested pacing (≈97 minutes of talking)

Block	Slides	Time	Natural break
Opening + Unit 01 · Language of Computers	1–6	≈24 min	—
Unit 02 · IP Fundamentals	7–10	≈23 min	end of session A
Unit 03 · Connectivity	11–15	≈18 min	—
Unit 04 · Operating Systems	16–18	≈12 min	end of session B
Unit 05 · Security Foundations	19–22	≈17 min	—
Closing	23	≈3 min	end of session C

Three 45–50 minute sessions work well: A = slides 1–10, B = slides 11–18, C = slides 19–23 plus lab time. Add buffer for the TRY IT moments — they run long when the class is having fun.

Room & props checklist

- Projector + whiteboard — the binary and subnetting math belongs on the board (slides 5 and 9).
- A Cat6 patch cable and a spare RJ45 plug to pass around (slide 12).
- A fiber patch lead if you can get one — caps on (slide 13).
- A machine with a terminal / PowerShell for the live ping ladder (slide 18).
- Packet Tracer and a Linux VM ready for the /27 and command-line drills (slides 9 and 18).
- Students' phones — they will check their own IP address (slides 8 and 10).
- Printed student handouts — the TRY IT and CHECK YOURSELF answers are in the back of theirs.

Slide 1 · Cover — Cybersecurity Foundations ≈2 min

> **SAY**
>
> Welcome. By the end of this course you will do something most people think is magic: you will read the language computers actually speak. Five units — how computers count, how devices find each other, how data physically travels, the software that runs it all, and how to protect it.
>
> No prior experience needed — we start at absolute zero. You are here because you like solving puzzles, and everything in this course is a puzzle with an answer.

LAND THIS — No prerequisites. Five units take you from zero to job-ready junior admin.

ASK THE CLASS — Quick show of hands: who has ever opened up a computer? Who has set up a Wi-Fi router at home? (Either way is fine — that is the point of the next five units.)

Slide 2 · Roadmap — Five Stops to Your First Admin Job ≈3 min

SAY

Here is the whole course on one slide. Unit 1: how computers count. Unit 2: how devices find each other. Unit 3: how the data physically travels. Unit 4: the software that runs everything. Unit 5: how attackers target you — and how to stop them.

Each unit is one tool on your belt. Watch how they connect: the number 255 from Unit 1 comes back as the biggest number in an IP address in Unit 2. Nothing here stands alone.

LAND THIS — Five connected stops — each one feeds the next.

ASK THE CLASS — Which stop sounds most interesting to you? (Quick poll — remember their answers and point back to them on the closing slide.)

Slide 3 · Unit 01 divider — The Language of Computers ≈1 min

SAY

Unit one: the language of computers. In the next three slides you will read binary and hex — the two things that scare people away from IT. Spoiler: they are easier than they look.

TRANSITION — Let us start with the only two things a computer truly knows.

Slide 4 · Computers Know Exactly Two Things ≈5 min

SAY

A computer is billions of microscopic switches. Each switch is ON or OFF — there is no "maybe". We write ON as 1 and OFF as 0. One switch is one bit. Eight bits make one byte.

And here is the entire trick of computing: every photo, song, password, and video game is just a very long pattern of these switches. Why only two states? Because a switch that only has to tell ON from OFF is cheap, fast, and almost impossible to confuse — a little electrical noise never turns a clear ON into an OFF.

LAND THIS — Bit = one switch. Byte = 8 bits. Everything digital is just patterns of them.

DO LIVE — Have eight students stand — arm up is 1, arm down is 0. The class just became a byte. Call out a pattern (1-0-1-0-1-1-0-0) and tell them they will read it as a number on the next slide.

WATCH FOR — Students assume files contain something more mysterious than numbers.

Kill it early: a text file is just bytes. There is nothing else in there.

Slide 5 · Reading Binary — Each Position Doubles ≈7 min

> **SAY**
>
> Reading binary is three steps. Write the place values: start at 1 on the right and double as you go left — 1, 2, 4, 8, 16, 32, 64, 128. Put the bits underneath. Then add only the places that have a 1.
>
> Watch: 10101100. That is 128, skip 64, 32, skip 16, 8, 4, skip the rest. 128 plus 32 plus 8 plus 4 equals 172. That is it — you just read machine language.

LAND THIS — Only the 1s count. Add their place values.

ASK THE CLASS — "What is the biggest number one byte can hold?" — 255, all eight switches on. Then say: remember 255; it comes back in Unit 2 as the biggest number in an IP address.

DO LIVE — Solve 10101100 on the board slowly, narrating every column. Then have everyone do the TRY IT (00001111) on paper — cold-call for the answer (15). Fast finishers get 00101010 (42).

WATCH FOR — The classic mistake: adding the place values of the 0 columns. Keep repeating "only the 1s count".

Slide 6 · ASCII & Hex — Letters Are Numbers in Costume ≈6 min

> **SAY**
>
> If everything is bytes, how does a computer store text? By agreement. ASCII gives every character a number: capital A is 65, small a is 97, even the space is 32. So the word "Hi" is two bytes: 72, then 105.
>
> Writing eight bits per letter gets old fast, so admins use hexadecimal: 0 to 9, then A to F. One hex digit covers exactly four bits, so one byte is always two hex digits. Our 172? Split 1010 1100 into two halves — A and C — so 172 is AC in hex.
>
> You will see hex every single day: MAC addresses like 7C:2E:BD:41:0A:9F, and error codes like 0x80070005 — that one is "access denied" in costume.

LAND THIS — ASCII turns letters into numbers; hex turns binary into shorthand.

ASK THE CLASS — "If A is 65, what number is B?" — 66. Let them enjoy how obvious it is; then ask what D is (68).

WATCH FOR — Hex panic. Emphasize it is not new math — it is the same bits, written

shorter.

Slide 7 · Unit 02 divider — IP Fundamentals ≈1 min

SAY

Unit two: IP addresses — how every device gets found on a network. This is the unit working admins use every single day, and it has the best party trick in the course: subnetting by hand.

TRANSITION — First, forget the numbers. An IP address is just a street address.

Slide 8 · An IP Address Is a Street Address ≈6 min

SAY

An IP address is a street address. 192.168.1.10 — the first part is your street, the network; the last part is your house number, the host. Each of the four numbers is one byte — an octet — which is why they max out at 255. Unit 1 paying off already.

Four settings get a device online: an IP address — who you are. A subnet mask — which network you belong to. A default gateway — the door out, your router. And DNS — the phone book that turns names into numbers. Miss any one of the four and something specific breaks.

LAND THIS — Network part = street, host part = house number. Four settings get you online.

ASK THE CLASS — "Who wants to check their phone's IP address right now?" (Settings → Wi-Fi → network details.) Read one out loud — it will almost certainly be 192.168.x.x or 10.x.x.x. Tell them to remember it for slide 10.

WATCH FOR — Mask anxiety. Defuse it: the mask is just the ruler that says where the street ends and the house begins.

Slide 9 · Worked Example — Subnetting 192.168.1.0/26 ≈10 min

SAY

Here is the party trick. You start with 192.168.1.0/24 — 256 addresses. Borrow 2 bits for the network and you get a /26: four subnets of 64 addresses each.

Four steps. One: the mask's last octet is 192. Two: block size is 256 minus 192 — that is 64. Three: count the networks up by 64 — 0, 64, 128, 192. Four: first host is network plus 1, broadcast is the next network minus 1, last host is broadcast minus 1.

Why minus two? The first address names the network itself; the last one shouts to everyone. Both are reserved — no device can have them.

LAND THIS — Block size = 256 − the mask's last octet. Everything else is +1 or −1.

ASK THE CLASS — "What is the first address called? The last?" — make the class answer the "why minus 2" themselves.

DO LIVE — Derive subnet 2's row on the board together (.64 / .65 / .126 / .127). Then set the /27 drill — block size 32, eight subnets — for the lab, where they verify it in Packet Tracer.

WATCH FOR — Students memorizing the table instead of the method. The method fits in one line; make them say it with you: "256 minus the mask's last octet."

Slide 10 · Private Addresses, NAT & IPv6 ≈6 min

SAY

Some addresses stay home. Three ranges — 10.x.x.x, 172.16 to 172.31, and 192.168.x.x — are free for anyone to use indoors, and they never appear on the public internet. Check your phone again: almost certainly one of those. So how are you on the internet?

NAT. Your router lets the whole house share one public IP address and keeps track of which reply belongs to which device. Bonus: it also hides your devices from the outside.

And the future: IPv4's 4.3 billion addresses ran out, so IPv6 stretches addresses to 128 bits, written in hex — which you can already read. 2001:db8::1 — the "::" just means "fill in zeros here".

LAND THIS — Private ranges stay indoors, NAT shares one public IP, IPv6 is the same idea with far more room.

ASK THE CLASS — Bonus challenge: "Who can compress 2001:0db8:0000:0000:0000:00ff:0000:0042?" (2001:db8::ff:0:42 — one zero-run becomes ::,

leading zeros drop.)

Slide 11 · Unit 03 divider — Connectivity ≈1 min

SAY

Unit three: how the bits physically move — electricity through copper, light through glass, or radio through air. By the end you will know exactly which cable to grab in any situation.

TRANSITION — We start with the workhorse you have all touched a hundred times.

Slide 12 · Copper — the Ethernet Workhorse ≈4 min

SAY

The cable at your desk is twisted-pair copper — Cat5e or Cat6 — with the chunky RJ45 plug. Bits travel as pulses of voltage. Ethernet — that is just the IEEE 802.3 committee's name — went from 10 megabits in 1983 to 400 gigabits in 2017. Forty thousand times faster; same idea.

One rule has never changed: 100 meters per run — past that, the signal fades. Copper's other two enemies: interference, because copper is an antenna, so keep it away from power cables and motors; and cheap crimps, which cause the weirdest network bugs you will ever chase. When in doubt, swap the cable first.

LAND THIS — Copper is the cheap default — and the 100 m rule never changes.

DO LIVE — Pass around a real Cat6 patch cable and a spare RJ45 plug. Let them feel the clip and look at the eight wires.

Slide 13 · Fiber — Data at the Speed of Light ≈5 min

SAY

A fiber cable is a hair-thin strand of glass. Light goes in one end and bounces off the walls all the way down — total internal reflection. Flash means 1, dark means 0. Bits again — just much, much faster.

Two flavors. Single-mode: one thin core, one straight beam — it goes kilometers, so that is ISP backbones and links between buildings. Multi-mode: a wider core with many bouncing beams — cheaper gear, good for hundreds of meters inside one building.

Admins love fiber: immune to electrical interference, insanely fast — and "fiber to the home" means the glass may already reach your house.

LAND THIS — Light in glass: kilometers, no interference.

ASK THE CLASS — "Copper or fiber next to a factory floor full of big electric motors?" — fiber, because it is immune to interference.

DO LIVE — If you have a fiber patch lead, pass it around (caps on). Point out how thin the core is compared to the copper cable from the last slide.

Slide 14 · Wireless — Wi-Fi Is Radio, Politely Sharing ≈5 min

SAY

Wi-Fi sends bits through the air as radio waves — IEEE 802.11. The catch: everybody shares the same air. Your laptop, the neighbor's router, Bluetooth gadgets, the microwave — all in the same invisible space.

Two bands. 2.4 GHz — the old faithful: it reaches far and through walls, but it is slow and crowded. 5 and 6 GHz — the fast lane: much faster with more room, but walls eat it. Same trade-off every time.

Three things kill your signal: walls and metal, channel crowding — on 2.4 GHz only channels 1, 6, and 11 do not overlap, so those are the only ones to pick — and distance.

LAND THIS — The eternal trade-off: range versus speed — and the air is shared.

ASK THE CLASS — "Your phone has one bar at the far end of the house — which band are you probably on?" — 2.4 GHz, because it reaches farther.

Slide 15 · The Decision — If It Doesn't Move, Wire It ≈3 min

SAY

The whole unit in one line: if it doesn't move, wire it. Copper for desks and printers. Fiber for distance — buildings and backbones. Wi-Fi for anything that moves — people, phones, gadgets.

Walk into any building now and you can explain every cable you see.

LAND THIS — Copper by default, fiber for distance, wireless for mobility.

ASK THE CLASS — Run the four scenarios aloud: a receptionist's desktop (copper) · two buildings 300 m apart (fiber) · warehouse barcode scanners (wireless) · the uplink from a switch to the server room (fiber). Cold-call each.

Slide 16 · Unit 04 divider — Operating Systems ≈1 min

SAY

Unit four: operating systems. Linux runs the servers, Windows runs the desks — and you are going to run both. Good news: they do the same jobs, just with different accents.

TRANSITION — First, what does an operating system actually do all day?

Slide 17 · The OS Is the Building Manager ≈5 min

SAY

Hardware is the building, apps are the tenants, and the operating system is the building manager. It manages four things: files — where data lives; users — who is allowed to do what; memory and CPU — which program gets attention; and the network — how data leaves the machine.

Linux: free and open source, runs most of the internet's servers, and you drive it with typed commands — "everything is a file." Windows: most office desktops, graphical first, PowerShell when you mean business, and Active Directory manages users company-wide.

Same jobs, two accents. A good admin is bilingual.

LAND THIS — The OS manages files, users, memory/CPU, and network — on both systems.

ASK THE CLASS — "Who has used Linux? Who has used Windows?" — then point out they have all used Linux indirectly: Android phones and most websites run on it.

Slide 18 · Three Jobs Every Admin Does Daily ≈6 min

SAY

Three jobs you will do every day. One: manage users with least privilege — every account gets only the access its job needs. The intern gets the team folder, not the finance drive.

Two: find your way from the command line. Clicking is fine once; typing is repeatable — a command can be saved and run on a hundred machines. pwd, ls, cd, ipconfig.

Three: patch. Most real-world hacks exploit bugs that already have a fix — an updated system is a boring target.

And your first admin command: ping 192.168.1.1. “Reply from…” means the road to your router is alive.

LAND THIS — Least privilege, the command line, and patching — daily, forever.

DO LIVE — Project a terminal or PowerShell window. Run the ping ladder live: ping the gateway, then 8.8.8.8, then google.com — narrate what each rung proves (machine → internet → DNS).

WATCH FOR — Terminal fear. Reframe it: typing is just clicking, but repeatable.

Slide 19 · Unit 05 divider — Security Foundations ≈1 min

SAY

Final unit: security. Here is the uncomfortable truth — the easiest attacks do not target machines. They target people. So this unit is mostly about habits, not hardware.

TRANSITION — We start with the lock on the front door: your password.

Slide 20 · Passwords — Length Beats Complexity ≈6 min

SAY

Attackers do not guess passwords by hand — computers try billions per second. Every extra character multiplies their work. Meanwhile, humans handle complexity rules predictably: capital first letter, number at the end, an @ for an a. Cracking tools learned those tricks years ago.

So modern advice flips the script: go long. coffee-river-paper-moon — 23 characters of plain words — beats Tr0ub4dor&3. Easy for your brain, because your brain keeps the story; brutal for machines, because of the length.

The NIST rulebook: make it long, make it unique per account, let a password manager do the remembering — and no forced monthly changes, because people just add a "1" and move on.

LAND THIS — Length beats complexity. Unique per account. Manager + MFA.

ASK THE CLASS — "Be honest — who reuses the same password everywhere?" (Most hands go up.) Perfect: "That is how one leak becomes ten."

Slide 21 · MFA & the Five Phishing Red Flags ≈6 min

SAY

One lock is not enough. Something you know — a password. Something you have — your phone or a security key. Something you are — your fingerprint. Any two of the three is multi-factor authentication: a leaked password alone gets an attacker nowhere.

Which is exactly why attackers phish instead. Five red flags. Urgency — "your account closes in 24 hours!" Almost-right sender — micr0soft with a zero. Weird links — hover first and read the real destination. Surprise attachments — invoice_2084.pdf.exe is not a PDF; the last extension is the truth. And asking for secrets — real IT never asks for your password or your MFA code. Ever.

LAND THIS — MFA stops leaked passwords; five red flags stop phish.

DO LIVE — Show the sample phish email from the handout's TRY IT box on screen. Have the class shout out each red flag as they spot it — all five are in there.

Slide 22 · Your First Week on the Job + Where This Leads ≈4 min

SAY

Your first week on any IT job: turn on automatic updates — every machine, including your phone. Lock every screen — Win+L until it is a reflex. Start a password manager and MFA, and protect your email first, because it resets everything else. Hover before you click. Ask "do they need this access?" for every account you create. And back up before you change anything — the undo button of real life.

Where does this course lead? The NICE framework maps the ladder: IT support, then system or network administrator, then security analyst. You are standing on rung one.

LAND THIS — Habits beat heroics.

Slide 23 · Closing — You Speak Machine Now ≈3 min

SAY

Five units ago you could not read binary. Now: computers are switches, and every file is a pattern of bits. An IP address is a street address, and masks decide your neighbors. Copper for desks, fiber for distance, Wi-Fi for freedom. The OS is the building manager — Linux and Windows, same jobs. And long passphrases plus MFA plus a skeptical eye stop most attacks.

You speak machine now. Keep practicing: Packet Tracer for the subnets, Wireshark to watch real packets, a Linux VM to break and fix — and read RFC 791, the original spec. It is short, and you will never forget that you read it. Welcome to IT.

LAND THIS — Recap the five units in one breath each — then point them at practice.

ASK THE CLASS — Closing round: "Which unit are you going to practice first?"